Workload Management: DIRAC and DiracX as compute backends

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ARM and GPUs: HPC and clouds see them first



276 / 500
TOP500 systems with GPUs, June 2026 (110 in 2018)

35
ARM systems in the TOP500 (5 in 2020)

Our communities
run mostly on grid sites, and use a few HPCs opportunistically.

2010
first GPU system at the top of the TOP500

2018
ARM in a large cloud (AWS)

2020
first ARM system at the top of the TOP500

2025
first ARM pledges in WLCG, ATLAS only

2026
GPUs still cannot be pledged in WLCG

From HPC to the grid, and to the WMS

Grid sites buy from the same market: "Hardware technology evolution now is dominated by the needs of AI" (A. Sciabà, CERN IT, July 2026).

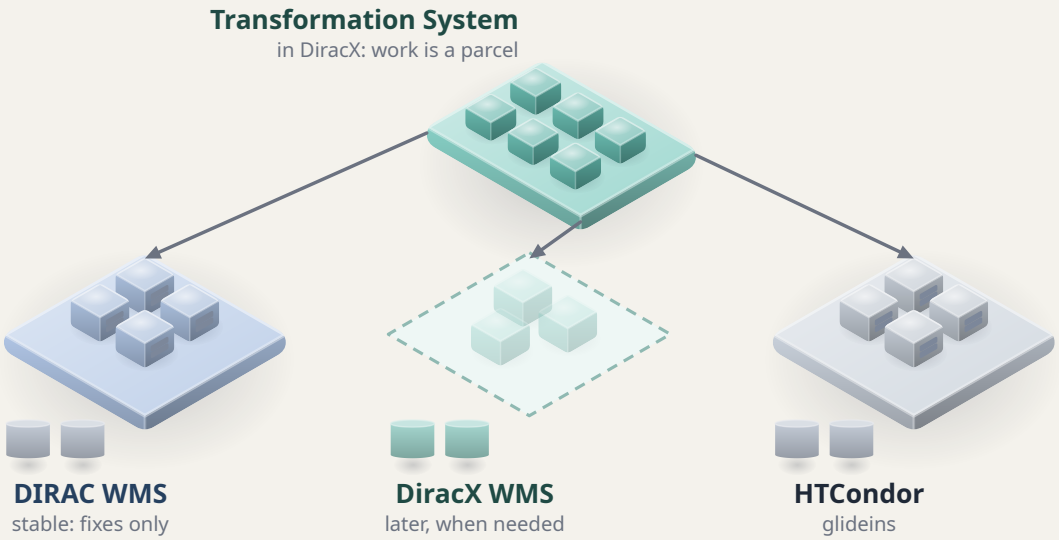
DRIVER, AND WHY	HPC AND CLOUDS FIRST	GRID SITES NOW	WHAT THE WMS MUST DO
AI demand Expected gains of $2.6–4.4T a year (McKinsey); NVIDIA data-centre revenue $194B in FY2026	GPUs in 276 of the TOP500	ATLAS: ~280 GPUs at 8 grid sites, little used; no GPU pledge before 2028	**Schedule CPU and GPU work together; match on capabilities**
Energy and hardware cost 2022 energy shock (CERN ran 20% less in 2023), climate targets, memory prices ×6 with AI	ARM in 35 of the TOP500; most new AWS CPU capacity is ARM	ATLAS: ARM up to half of a pledge, 11k slots at 5 sites; Glasgow ~20% ARM	**One site, two architectures: images and matching per architecture**
More cores per node Clock speed stalled around 2005 (power wall): extra transistors became cores. 32 per socket in 2017, 192 in 2024	HPCs hand out whole nodes: 112 cores on MareNostrum 5	Multicore pilots: 8 cores for 48 h (CMS)	**Fill nodes, not slots: multicore payloads, bundling, memory-aware matching**

In DIRAC: temporary solutions, to keep it stable. **In DiracX**: the long-term solutions, designed for this.

1 · Starting from the transformation system

What it changes for workload management

The transformation system sees compute backends



each backend keeps its own data and its own data model

The transformation system (Tuesday's talks) sends parcels to compute backends. **The DIRAC WMS is one of them**, with its own data, as it is.

What separate backends give us

A stable DIRAC

The DIRAC WMS keeps its data and its behaviour. No large change: fixes and performance only, until communities move away from it.

A DiracX free of the legacy model

DiracX does not inherit DIRAC's jobs, tables or states. Work is a **parcel**; the WMS behind it is designed for it.

No rush for a DiracX WMS

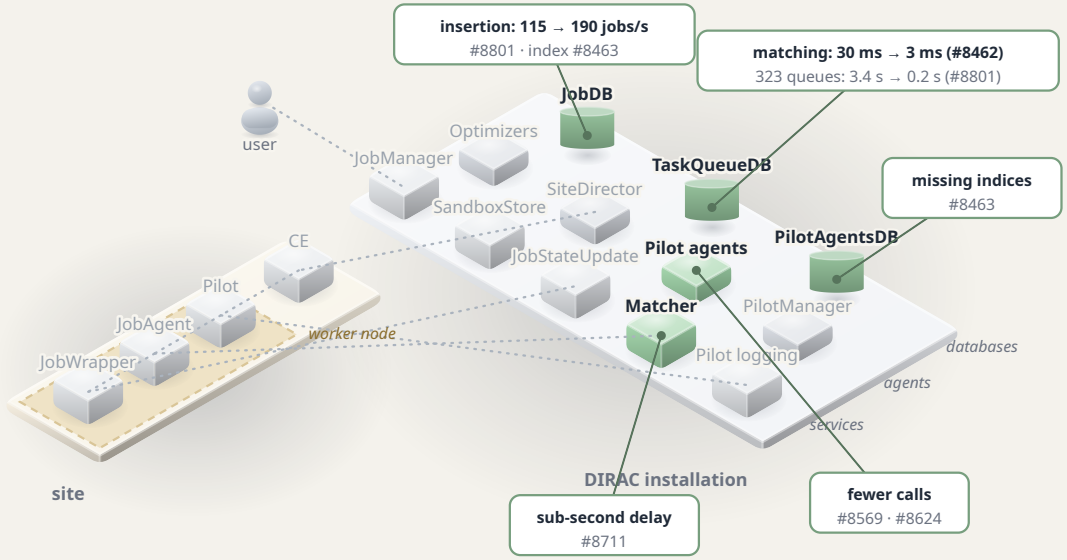
Most communities can rely on their existing WMS for now: DIRAC, or their own (see the CMS talk just before).

> The DiracX WMS is not a priority. It will move slowly, in parallel, and get a plan when the transformation system needs a native backend.

2 · DIRAC: fixes and performance

What keeps it running while we migrate

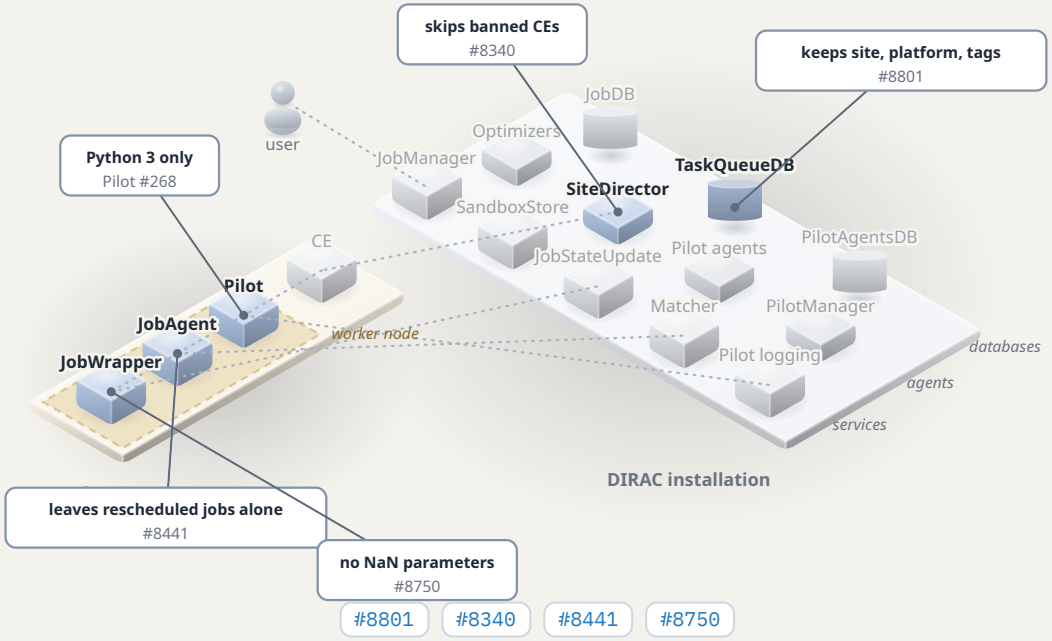
Faster



#8462 #8801 #8711 #8463 #8569 #8624 #8524

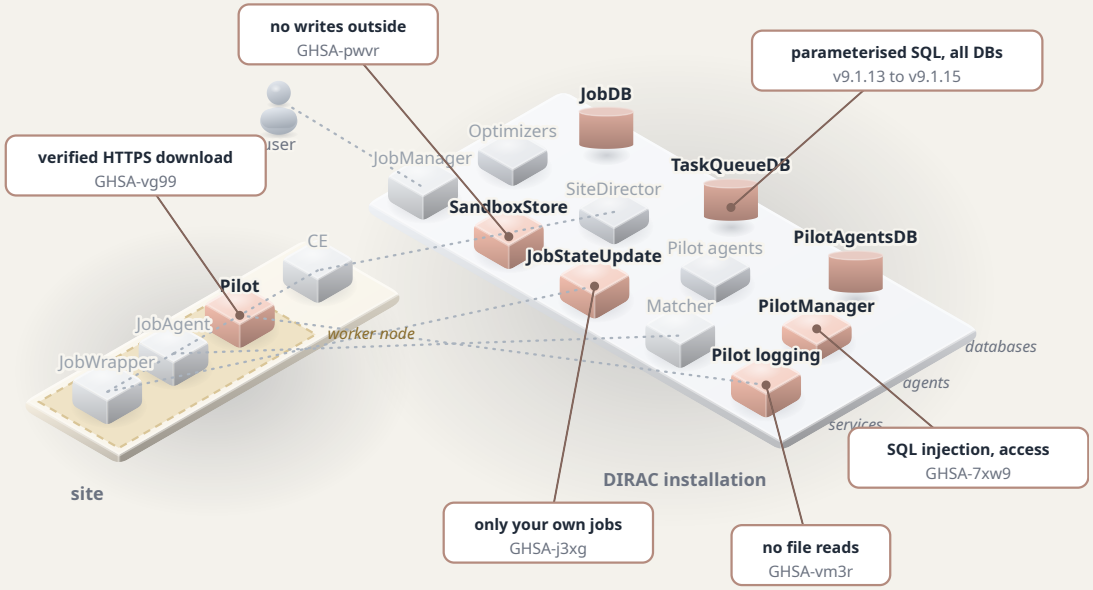
Several fixes made the DIRAC WMS faster, for installations at v9 and above. Most of these numbers need v9.1.

More reliable



> None of these change how DIRAC works. They remove ways it could fail.

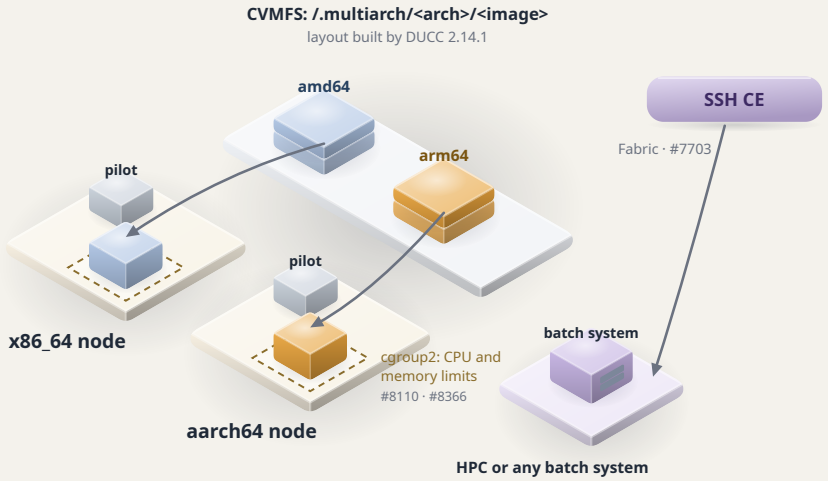
Safer



GHSA-j3xg GHSA-pwvr GHSA-vm3r fixed in v9.1.20 · v9.0.26 · v8.0.85 GHSA-7xw9 GHSA-vg99 fixed in v9.1.10 · v9.0.22

→ *DIRAC is patched, not developed:* **upgrading is how you get these**

Heterogeneous nodes



the pilot reads uname -m and picks the matching directory (#8456)

- **aarch64 images** `DUW11`: #8456, open, to certify. Needs CVMFS 2.14 (DUCC 2.14.1) where images are published; any client reads it.
- **Limits** `DUW11` and **HPC** `DUW11`: in v9.0.
- **Not solved in DIRAC**: a job for x86_64 *or* aarch64.

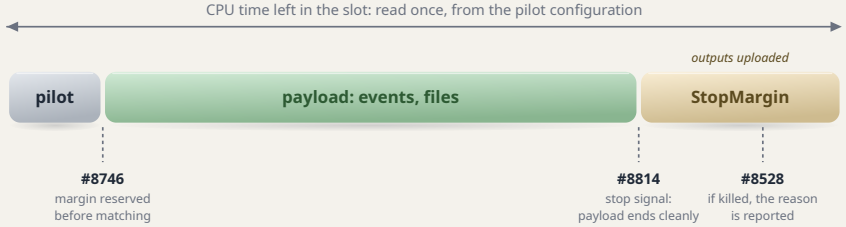
Time left in the slot

The problem

- Jobs in containers could not ask the batch system for the time left.
- The post-processing margin was known only to the Watchdog, which could only kill.

An example: an MC job sized for 183 events produced 150, was killed, and its output was never uploaded.

Since v9.1.4 (#8479): the pilot reads the time left once; jobs read it from the pilot configuration.

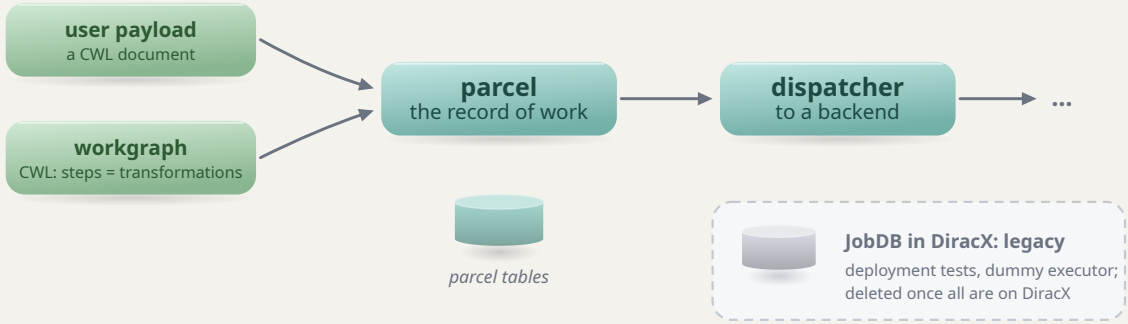


> The three fixes above are in **v9.1.21** only; see "Allocated time management" in the Administrator Guide. CPU work is normalised with DB12: next talk.

3 · DiracX: parcels, not jobs

And the proofs of concept around the future WMS

No jobs in DiracX



- A user payload is a parcel, like a bundle of transformation inputs. A failed parcel is final: resubmitting creates a new one.
- We do not port job monitoring or the job manager `DUW11`.

- The JobDB in DiracX is legacy. It receives job status updates from DIRAC, to **test that a DiracX deployment works**.
- It also feeds the **dummy job executor** (diracx#995): jobs go `Waiting → Matched → Running → Done` without running anything. A stand-in backend, to test the transformation system end to end.
- **Once everyone is on DiracX, it is deleted.**

CWL describes all the work

COMMON WORKFLOW LANGUAGE

One format (DX-ADR-007, draft): a user payload, a workgraph and each transformation are CWL v1.2 documents.

- DiracX settings are `dirac:` **hints**: every document stays plain, runnable CWL.
- The same document runs with `cwltool`, locally, and at scale.
- Processes are resolved before dispatch: the node fetches no definitions.

For a WMS backend

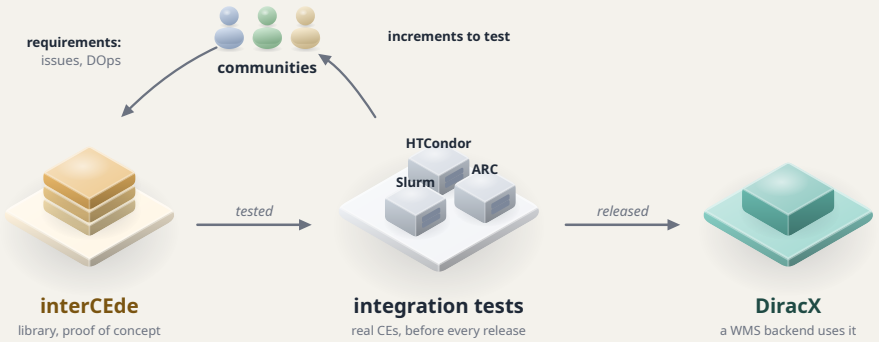
- Matching requirements come from the hints.
- The job wrapper resolves inputs (`lfn:` , `sandbox:`) and reports a status per input.
- DiracX registers the outputs after the parcel ends.

Yesterday: hands-on sessions to convert DIRAC workflows to CWL.

With the CWL community: cwltool#2302 path mapper, cwltool#2166 Apptainer sandboxes, schema_salad#1064: merged. cwl-utils#446 QuickJS engine: in review.

interCEde: computing elements, explored from use cases

- **An exploratory project**, from the use cases DIRAC covers, and those it does not but should.
- **Its own repository**: integration tests against real CEs, so that a change cannot break a production instance.
- **Ends with ADRs and a first implementation**; the ADRs are merged once that implementation works.



Open call: if you work with DIRAC computing elements, test the increments of interCEde and tell us what does not fit your needs, or what DIRAC does not cover: in an issue, or at a DOps.

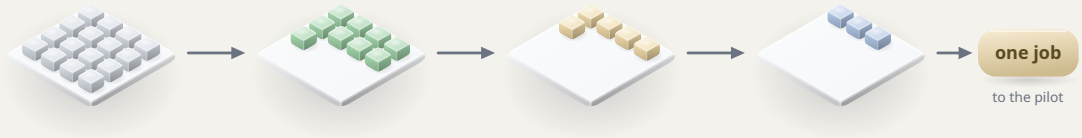
A matcher for the DiracX WMS

Why

- DIRAC's matcher will not be fast enough for what is coming for some communities, such as LHCb.
- It cannot describe heterogeneous needs: a payload that runs on GPU A, *or* GPU B, *or* CPU C.

The proof of concept (poc-dirac-job-matchmaking)

- A deliberately simple Python matcher, and load tests on LHCb-shaped data to compare solutions.
- The solution must be easy to deploy and to rebuild: chaos tests with Redis switched off.



one request at a time, in Python: every waiting job is checked

Next: matching in Redis, measured against this baseline. Then an ADR, then the implementation in DiracX.

Pilot management: port paused

We started

Porting pilot management from DIRAC to DiracX as it is: a pilot database and router landed (diracx#570) DUW11.

We learned

The data model could be much better. Changing it would mean large changes in DIRAC, which we keep stable.

So we paused

Registration, pilot tokens and logging wait for the DiracX WMS backend. The re-thinking has not started.

> Pilots belong to the DiracX WMS backend, which comes after the transformation system. The DIRAC pilot keeps being patched.

The job wrapper: one for all of us

Today there are several

- DIRAC's JobWrapper: pre-processing, payload, post-processing; an offline variant for push.
- The `dirac-cwl` wrapper, experimental in DIRAC.
- Executor prototypes in DiracX.

Each community also adds its own steps around the payload.

Requirements to gather

- Inputs: data, sandboxes, protocol access.
- Time left, graceful stop, containers, architectures.
- Nodes without outbound network.
- Reporting: status, parameters, logs.
- What happens when an upload fails.

We are looking for someone to gather the requirements and the feedback, build a proof of concept, then design the ADR, working alongside someone who has been through the process once. Come and talk to us this week.

Take-aways

THE MODEL

- The transformation system sends parcels to compute backends; the DIRAC WMS is one of them.
- DIRAC stays stable; DiracX is free of DIRAC's job model.

FOR INSTALLATIONS

- Upgrade to v9.1: faster matching and job insertion, time-left fixes (v9.1.21), security patches.
- Multi-architecture images: #8456, soon.

FOR DIRACX

- No jobs: parcels. The JobDB in DiracX only tests deployments, then goes.
- The DiracX WMS is not a priority; it will get a plan when needed.
- CWL describes everything that runs.
- interCEde and the matcher: exploratory, with tests; ADRs after a first implementation. Pilot management: paused.

WE NEED YOU

- Someone to lead the job wrapper requirements and proof of concept; your requirements for interCEde and the matcher.

Thank you

Questions and comments welcome, this week and after.

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